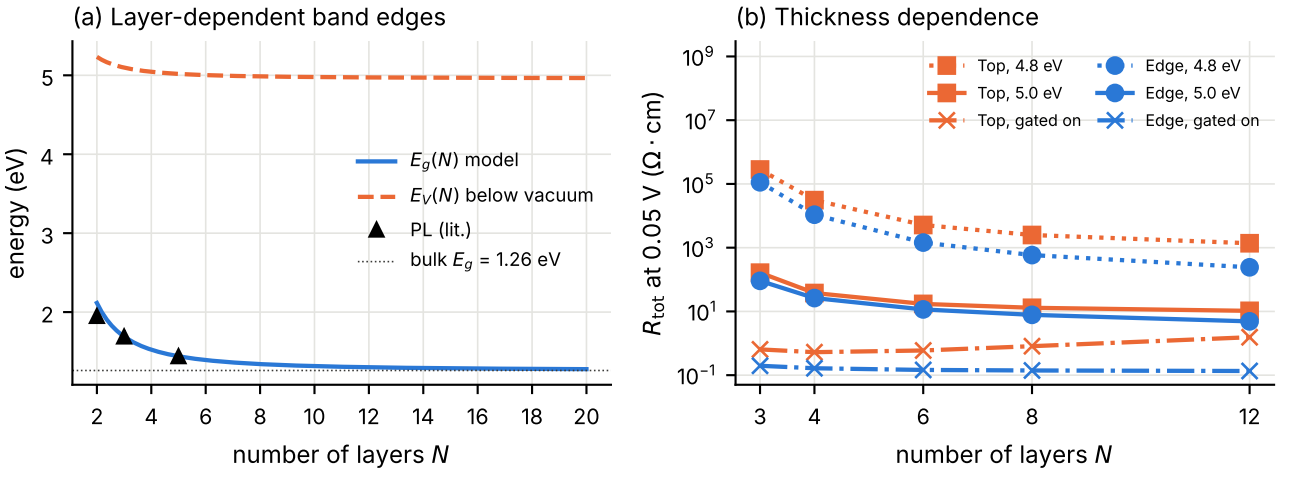
(a) Layer-dependent band edges
energy (eV)
number of layers N
$E_g(N)$ model
$E_V(N)$ below vacuum
PL (lit.)
bulk E_g = 1.26 eV
(b) Thickness dependence
R_{tot} at 0.05 V (Ω · cm)
number of layers N
Top, 4.8 eV
Top, 5.0 eV
Top, gated on
Edge, 4.8 eV
Edge, 5.0 eV
Edge, gated on